1. Introduction

The purpose of our Entity Relationship diagram is to display the items and relationships in a real-world library. This is a continuation of the previous discussions and projects created for this course. We represent a library with a Entity Relationship diagram so that we can easily see what different items could reside in a library as well as how they can interact with one another. In order to properly read the diagram it can be important to know the following acronyms and terms: ID for Identification Number, N to represent any number of items, or M for the same purpose. Admin, a super user that controls others accounts or has otherwise elevated permissions.

2. ER Modeling Components:

Entities:

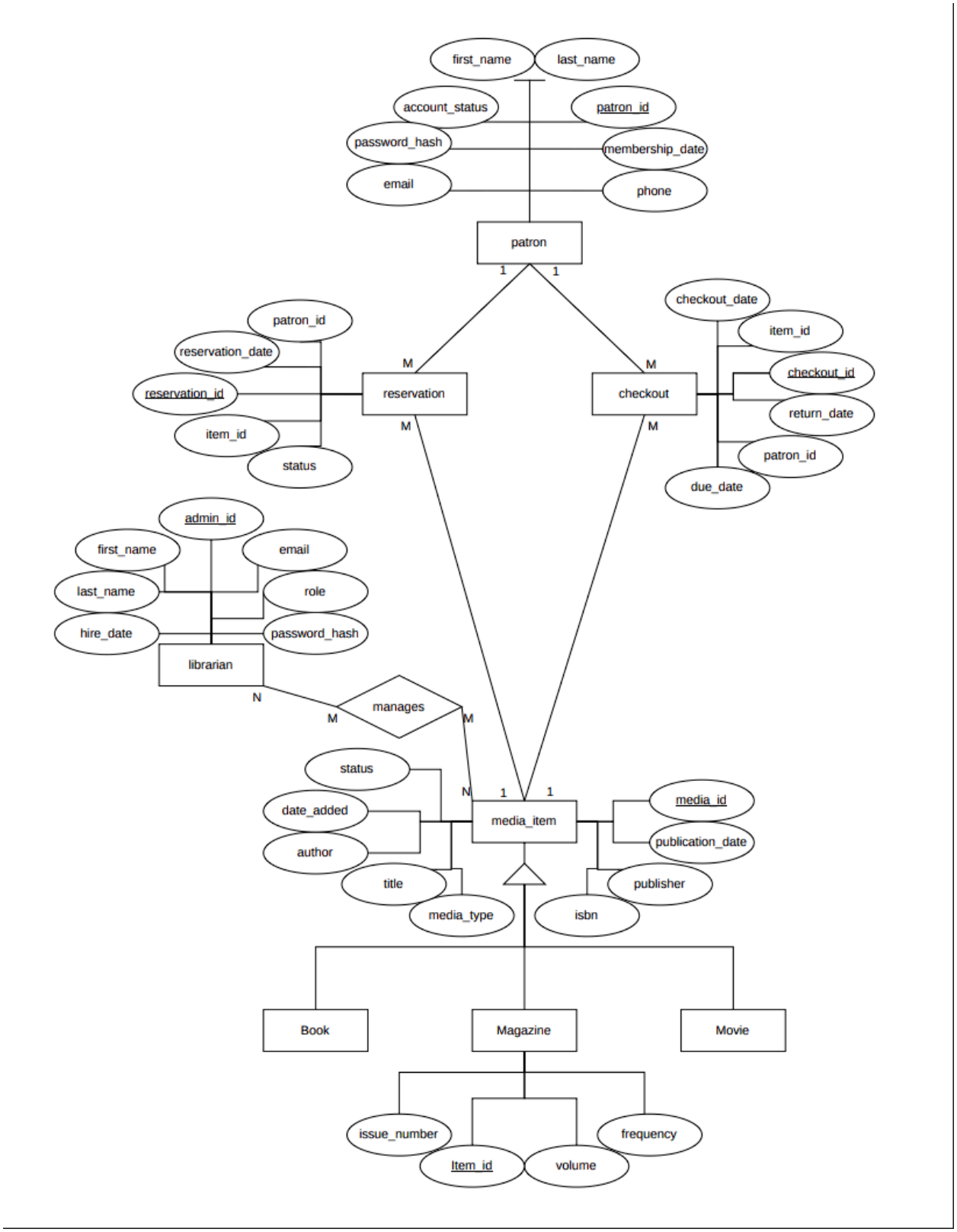
- Patron
    - Patron_id
    - Email
    - Password_hash
    - Account_status
    - First_name
    - Last_name
    - Membership_date
    - Phone
- Librarian
    - Admin_id
    - Password_hash
    - Role
    - Email
    - First_name
    - Last_name
    - Hire_date
- Book
- Magazine
- Movie
- Media_item
    - Media_id
    - Status
    - Date_added
    - Author
    - Title
    - Media_type
    - Isbn
    - Publisher
    - Publication_date
- Checkout
    - Checkout_id

        - Checkout_date
        - Item_id
        - Return_date
        - Patron_id
        - Due_date
- Reservation
    - Reservation_id
    - Item_id
    - Patron_id
    - Reservation_date
    - Status

Relations:

- A patron may make many reservations, but each reservation is connected with only one patron via patron_id.
- A patron may make many checkouts, but each checkout is connected with only one patron via patron_id.
- A media item may be checked out many times, but each checkout record only has one media item connected via item_id.
- A media item may be reserved many times, but each reservation record only has one media item connected via item_id.
- Many librarians manage many media_items.
- Book, magazine, and movie and specializations of media_item.

3. ER Model


first_name
last_name
account_status
patron_id
password_hash
membership_date
email
phone
patron
1
1
patron_id
reservation_date
reservation_id
item_id
status
M
M
reservation
checkout
M
M
checkout_date
item_id
checkout_id
return_date
patron_id
due_date
admin_id
first_name
email
last_name
role
hire_date
password_hash
librarian
N
manages
M
M
status
date_added
author
title
media_type
N
1
1
media_item
media_id
publication_date
publisher
isbn
Book
Magazine
Movie
issue_number
frequency
Item_id
volume